

Revised Generation Interconnection System Impact Study Report

For Queue Project Q23-2286

Indian Mesa 138 kV
December 21, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2286. The Interconnection Customer proposes a 59.1 MW solar photovoltaic facility interconnecting at the Indian Mesa 138 kV substation in the New Jersey area. The request was received on February 04, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 749 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,226,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: COW CRK – MARION 138 KV
Contingency: CAPRI – ROXBORO PLAN STA 138 KV Line
Loading: 115.3% (Rating: 185 MVA)
DFAX: 7.1%
Pre-project loading: 113.0%
2. Facility: ROXBORO PLANT STATION – INDIAN MESA 138 KV L
Contingency: ATHENIA – DAM HS 138 KV L
Loading: 108.2% (Rating: 201 MVA)
DFAX: 30.6%
Pre-project loading: 99.2%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
COW CRK SUBSTATION – MARION 138 KV line	Rebuild line with higher-capacity conductor	\$1,187,000	100% (\$1,187,000)
ROXBORO PLANT STATION – INDIAN MESA 138 KV L	Reconductor line	\$230,000	17% (\$39,000)
	Total allocated to project		\$1,226,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.